CAI BIRCH

Java Developer | Spring Boot, REST APIs and cloud deployment

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PROFILE

Java developer specialising in Spring Boot back-end services, with hands-on experience building and deploying microservices in Java and Kotlin with Docker, Kubernetes and CI/CD in an agile health-tech team, plus three years of earlier software development in Python and Matlab in research.

Built and run Skepparklar, a production e-learning platform with a Java 21 and Spring Boot 4 REST API, a React and TypeScript front end, Stripe payments and automated deployment to Linux. Responsible for the full lifecycle, from schema design to production support.

Years of running university lab classes have made me comfortable explaining technical decisions to people who aren't developers.

TECHNICAL SKILLS

- **Java and Spring:** Java 21, Kotlin, Spring Boot (v2/v3/v4), Spring Data JPA/JDBC, Hibernate
- **APIs and services:** RESTful APIs, microservices, Stripe webhooks, OAuth 2.0/OIDC (Keycloak, Auth0)
- **Cloud and deployment:** Docker, Kubernetes, CI/CD with GitHub Actions, Linux, nginx, systemd, Cloudflare
- **Databases:** PostgreSQL, SQL, schema design, Flyway migrations
- **Front end:** JavaScript, TypeScript, HTML, React, Vue.js, Tailwind CSS
- **Testing:** JUnit 5, Mockito, Testcontainers, Vitest, React Testing Library, pytest
- **Other languages:** Python, C#
- **Ways of working:** Agile/Scrum, Git, JIRA, code review, architecture decision records
- **Languages:** English (native) and Swedish (fluent)

EXPERIENCE

Software Developer (intern) | Braive AB — Stockholm | 2026

- Built and deployed microservices in Java and Kotlin with Spring Boot, plus admin interfaces in Vue.js, in an agile team using Git, JIRA and code review.
- Containerised services with Docker and worked in Kubernetes-based CI/CD pipelines from development through to production.
- Found and fixed race conditions in the backend logging flows, which made logging more reliable and protected data integrity.
- Improved how sensitive healthcare data was handled in PostgreSQL and Hibernate/JPA, applying security principles to cut the risk of manual errors.

Software Developer (intern) | Omika Health AB — Stockholm | 2025

- Designed and built a production-ready peer-to-peer video chat with WebRTC in React Native, including the signalling server and secure authentication.
- Built and debugged low-latency real-time flows and covered complex client interactions with tests.

Prison Officer (part-time and summer cover) | Swedish Prison and Probation Service — Stockholm | until August 2026

- Worked in a security-classified government environment with strict confidentiality requirements. Passed a security vetting under the Swedish Protective Security Act.

Lab Teacher and Researcher | Loughborough University — UK | 2018-2021

- Wrote Python and Matlab software for biomechanical modelling, simulation and analysis of sensor and motion data with OpenSim.
- Built and debugged tools and data collection pipelines for lab classes of more than 100 students, and mentored colleagues in cross-disciplinary teams.

OWN PRODUCT: SKEPPARKLAR (WWW.SKEPPARKLAR.SE)

Java 21, Spring Boot 4, PostgreSQL, React 19, TypeScript, Stripe, GitHub Actions

- Designed and built a full-stack e-learning platform for Swedish boating licence courses, from database schema to production. About 46,000 lines of code and 1,200 commits.
- Wrote the REST API in Java 21 and Spring Boot 4 with Spring Data JDBC, PostgreSQL and 15 versioned Flyway migrations.
- Deploy through GitHub Actions to a Linux server behind Cloudflare and nginx, run by systemd with automatic database backups.
- Built my own authentication with no external provider: Argon2 password hashing, TOTP two-factor login, email verification, password reset, per-IP rate limiting and sessions stored in PostgreSQL with Spring Session.
- Integrated Stripe payments with webhooks, refunds and course access tied to purchases.
- Tested the backend with JUnit 5 and Testcontainers against a real PostgreSQL database, and the front end with Vitest and React Testing Library.

EDUCATION

Higher Vocational Diploma (YH), Software Development with Java | YH Akademin, Sundsvall — 2024-2026

Two-year full-time programme focused on Java back-end development: OOP, data structures, algorithms, design patterns, Spring Boot, REST, microservices, Hibernate/JPA, SQL, Docker, CI/CD and Git. Two internships, at Braive AB and Omika Health AB.

MSc in Sport Science | Swedish School of Sport and Health Sciences (GIH), Stockholm

Biomechanics and motion analysis with signal processing, statistics and modelling in Matlab.

Research project in neuroscience | Lund University

BSc in Sport Science | Bangor University, UK

ACHIEVEMENTS

- 3rd place, Innovation Pioneers Hackathon 2025: built a prototype AI collaboration app in 48 hours.
- Attended Jfokus 2026, one of Europe's larger Java developer conferences.
- Associate Fellow of Advance HE, a recognised teaching qualification from Loughborough.
- GitHub: SchoolProject_API (Spring Boot REST API) and www.caibirch.com, a Three.js portfolio with Polly, an AI parrot running on a local LLM.

References available on request.